

1. Enumeration of scanme.nmap.org

Objective

The objective is to perform an initial security assessment of the publicly accessible host scanme.nmap.org using Nmap and identify:

- Host availability
- Open TCP ports
- Running services

Step 1: Host Discovery

Command

`nmap -sn scanme.nmap.org`

```
Session Actions Edit View Help
rah: corrupt history file /home/kali/.ssh_history
[~]kali@kali:~$ nmap -sn scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:16 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 8.34 seconds

[~]kali@kali:~$ nmap -sn scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:17 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 184.31 seconds

[~]kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:19 EDT
Stats: 8:08:05 elapsed: 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 104.00s done; ETC: 00:59 (4:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

[~]kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:01 EDT
Stats: 8:08:05 elapsed: 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 104.00s done; ETC: 00:59 (4:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 80.24 seconds

[~]kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:03 EDT
Stats: 8:08:05 elapsed: 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 104.00s done; ETC: 00:59 (4:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 23.26 seconds
```

Step 2: TCP Port Scanning

`nmap -sT scanme.nmap.org`

```
Session Actions Edit View Help
5861/tcp open  sip-tls
6880/tcp closed http-proxy
6881/tcp closed blackice-icecap

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

[~]kali@kali:~$ nmap -sT scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:01 EDT
Stats: 8:08:05 elapsed: 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 104.00s done; ETC: 00:59 (4:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 80.24 seconds

[~]kali@kali:~$ nmap -sT scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:03 EDT
Stats: 8:08:05 elapsed: 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 104.00s done; ETC: 00:59 (4:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0001s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 23.26 seconds
```

Step 3: Service Version Detection

```
nmap -sV scanme.nmap.org
```

The -sV option attempts to determine:

- Service name
- Application
- Version

Analysis

SSH may be required for legitimate remote administration. However, publicly exposing SSH increases the possibility of:

- Password-guessing attempts
- Credential attacks
- Exploitation of vulnerable SSH versions
- User enumeration depending on configuration

HTTP is normally required if the host provides a publicly accessible website.

Step 4: OS Fingerprinting

```
sudo nmap -O scanme.nmap.org
```

Or:

```
nmap -O -sV scanme.nmap.org
```

Step 5: Default Script Scanning

```
nmap -sC scanme.nmap.org
```

A combined scan can be performed as:

```
nmap -sC -sV -O scanme.nmap.org
```

For a broader TCP assessment:

```
nmap -p- -sC -sV scanme.nmap.org
```

-p- scans all TCP ports.

Recommendations

1. Disable unnecessary public services.

2. Restrict SSH using firewall rules.
3. Use SSH keys instead of passwords where possible.
4. Implement MFA for administrative access.
5. Keep operating systems and services patched.
6. Use HTTPS instead of unencrypted HTTP where applicable.
7. Monitor authentication logs.
8. Implement rate limiting and intrusion detection.
9. Minimize service-version disclosure.
10. Regularly perform vulnerability assessments.

Conclusion

Nmap enumeration provides valuable information about the host's externally visible attack surface. The discovered services should be compared with their business requirements, and services that are not required for public access should be disabled or restricted.

2. Enumeration of Metasploitable 2 — 192.168.56.101

Objective

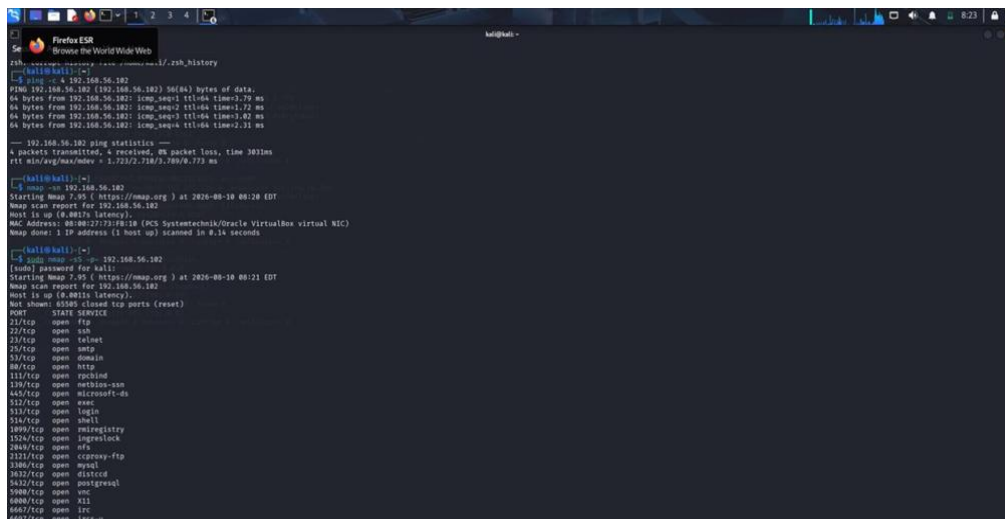
The purpose is to comprehensively enumerate the deliberately vulnerable Metasploitable 2 Linux server in an isolated laboratory.

Step 1: Host Discovery

```
nmap -sn 192.168.56.101
```

Step 2: Scan All TCP Ports

```
nmap -p- 192.168.56.101
```



```
kali@kali:~$ nmap -p- 192.168.56.101
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
135/tcp   open  msrpcs-svc
445/tcp   open  microsoftsds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1390/tcp  open  mlmregistry
1526/tcp  open  ingreslock
2800/tcp  open  nfs
2121/tcp  open  ciproxy-ftp
3800/tcp  open  mysql
5037/tcp  open  distdcd
5632/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6607/tcp  open  irc
6607/tcp  open  irc-u
```


SMB/NetBIOS — Ports 139/445

These services provide network file sharing and related functionality.

Potential risks include:

- Unauthorized file access
- Information disclosure
- Weak credentials
- Vulnerable SMB implementations

Risk: High.

Database Services

Examples include MySQL and PostgreSQL.

A database should generally not be directly accessible from untrusted networks.

Risk: High if externally accessible and weakly configured.

Risk Classification

High Risk

Services that may provide significant attack opportunities:

- Telnet
- Vulnerable FTP
- SMB
- Vulnerable HTTP applications
- Exposed databases
- Vulnerable RPC services

Medium Risk

- SSH
- SMTP
- DNS
- Other services requiring authentication/configuration review

Low Risk

A service can be classified as low risk when:

- It is necessary
- Properly configured
- Patched
- Restricted to trusted hosts

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:73:fb:10
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe73:fb10/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:57 errors:0 dropped:0 overruns:0 frame:0
          TX packets:72 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8714 (8.5 KB)  TX bytes:6464 (6.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$
```

Recommendations

For a production Linux server:

1. Disable Telnet.
2. Restrict FTP or replace it with SFTP.
3. Patch all services.
4. Restrict database ports.
5. Disable unnecessary RPC services.

Conclusion

Metasploitable 2 intentionally exposes many vulnerable services. Nmap provides the reconnaissance required to map these services and prioritize potential attack vectors in a controlled penetration-testing environment.

3. Windows Workstation — 192.168.56.102

Objective

The objective is to identify active services on the Windows workstation and determine whether they follow the principle of least privilege.

Step 1: Host Discovery

```
nmap -sn 192.168.56.102
```



```
sudo nmap -p- -sC -sV -O 192.168.56.102 -oN windows_scan.txt
```

Lateral Movement

Exposed SMB, RDP, and Windows remote-management services can become important after an attacker obtains valid credentials on another machine.

Unauthorized Access

Weak authentication combined with exposed remote-management services increases risk.

Hardening Recommendations

1. Disable unnecessary services

Unused SMB → Disable/restrict

Unused RDP → Disable

Unused WinRM → Disable

Unused RPC exposure → Restrict

2. Firewall

Allow management services only from trusted administrative systems.

3. Strong Authentication

Use:

- Strong passwords
- MFA where supported
- Account lockout/rate limiting

4. Patch Management

Regularly update:

- Windows
- SMB components
- RDP
- Management services
- Installed applications

5. Network Segmentation

Place administrative services on a restricted management network.

Nmap enumeration helps determine whether a Windows workstation exposes services beyond what is required. Reducing unnecessary services and restricting management protocols significantly decreases the attack surface.

Objective

OWASP Juice Shop is intentionally designed as a vulnerable web application for security training.

```
nmap -sn 192.168.56.103
```

```
nmap -p- 192.168.56.103
```



```
nmap -sV 192.168.56.10
```

Step 4: OS Fingerprinting

```
sudo nmap -O 192.168.56.103
```

```
Session + Terminal Emulator
Use the commandline

lnet
valid_lft forever preferred_lft forever
lnetk 192.168.0.0/24::192.168.0.0/24 scope link noprefixroute
valid_lft forever preferred_lft forever

--(kali@kali)--
$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.176 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.490 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.438 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 380.0ms
rtt min/avg/max/mdev = 0.036/0.092/0.176/0.051 ms

--(kali@kali)--
$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 18:23 EDT
Nmap scan report for 192.168.56.103
Host is up (0.000017s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
3000/tcp   filtered  xpp

Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds

--(kali@kali)--
$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 18:24 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0000090s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
3000/tcp   filtered  xpp

Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds

--(kali@kali)--
$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 18:24 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0000090s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
3000/tcp   filtered  xpp

Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds

--(kali@kali)--
$ sudo nmap -p80 -sV 192.168.56.103
```

```
Session + Terminal Emulator
Use the commandline

lnet
valid_lft forever preferred_lft forever
lnetk 192.168.0.0/24::192.168.0.0/24 scope link noprefixroute
valid_lft forever preferred_lft forever

--(kali@kali)--
$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.176 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.490 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.438 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 380.0ms
rtt min/avg/max/mdev = 0.036/0.092/0.176/0.051 ms

--(kali@kali)--
$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 18:23 EDT
Nmap scan report for 192.168.56.103
Host is up (0.000017s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
3000/tcp   filtered  xpp

Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds

--(kali@kali)--
$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 18:24 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0000090s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
3000/tcp   filtered  xpp

Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds

--(kali@kali)--
$ nmap -sn 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 18:24 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0000090s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
3000/tcp   filtered  xpp

Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds

--(kali@kali)--
$ sudo nmap -p80 -sV 192.168.56.103
```

Step 5: Default Scripts

`nmap -sC -sV 192.168.56.103`

Step 6: Web Service Identification

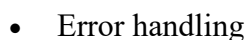
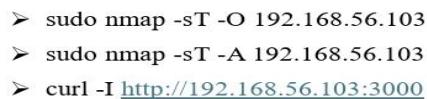
If port 80 is open:

`nmap -p 80 --script http-title,http-headers 192.168.56.103`

Web Application Exposure

A publicly accessible web service increases the attack surface because attackers can test:

- Authentication
- Authorization
- Input validation
- File handling
- Database interaction



Recommendations

1. Expose only required web ports.
2. Disable unnecessary services.
3. Restrict administrative interfaces.
4. Use HTTPS.

Nmap provides the network-level view required before application penetration testing. Understanding the exposed ports, services, and supporting infrastructure allows testers to prioritize web, API, authentication, and configuration testing.

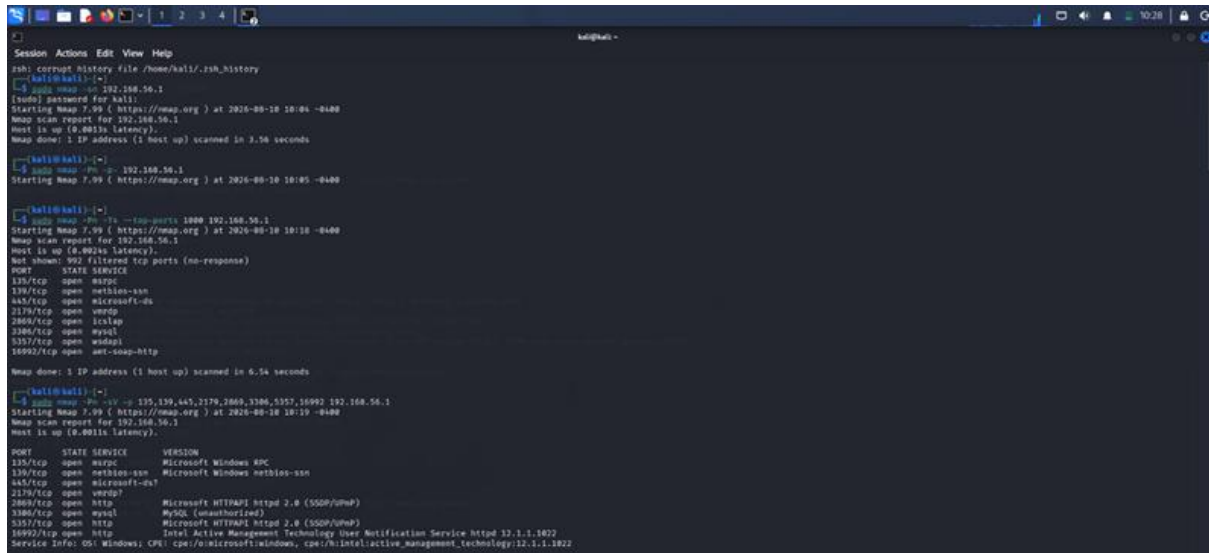
Objective

Step 1: Host Discovery

nmap -sn 192.168.56.

Step 2: TCP Port Scan

nmap -p- 192.168.56.



```
Session Actions Edit View Help
[sh] corrupt history file /home/kali/.ssh_history
--(kali@kali)~
[~] sudo nmap -p- 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:06 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).
Nmap done: 1 IP address (1 host up) scanned in 3.56 seconds

--(kali@kali)~
[~] sudo nmap -p- -P- 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:07 -0400

--(kali@kali)~
[~] sudo nmap -p- -T5 --top-ports 1000 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:10 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0024s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  murpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
2179/tcp  open  vncpp?
2869/tcp  open  iclslap
3306/tcp  open  mysql
5357/tcp  open  wsdapi
10992/tcp open  anti-snap-http

Nmap done: 1 IP address (1 host up) scanned in 6.54 seconds

--(kali@kali)~
[~] sudo nmap -p- -sV -P- -- 135,139,445,2179,2869,3306,5357,10992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).
PORT      STATE SERVICE        VERSION
135/tcp   open  murpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?  Microsoft Windows netbios-ssn
2179/tcp  open  vncpp?        Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
2869/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp  open  mysql         MySQL (unauthorized)
5357/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
10992/tcp open  http          Intel Active Management Technology User Notification Service httpd 12.1.1.1822
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1822

Nmap done: 1 IP address (1 host up) scanned in 6.54 seconds
```

Step 3: Service Detection

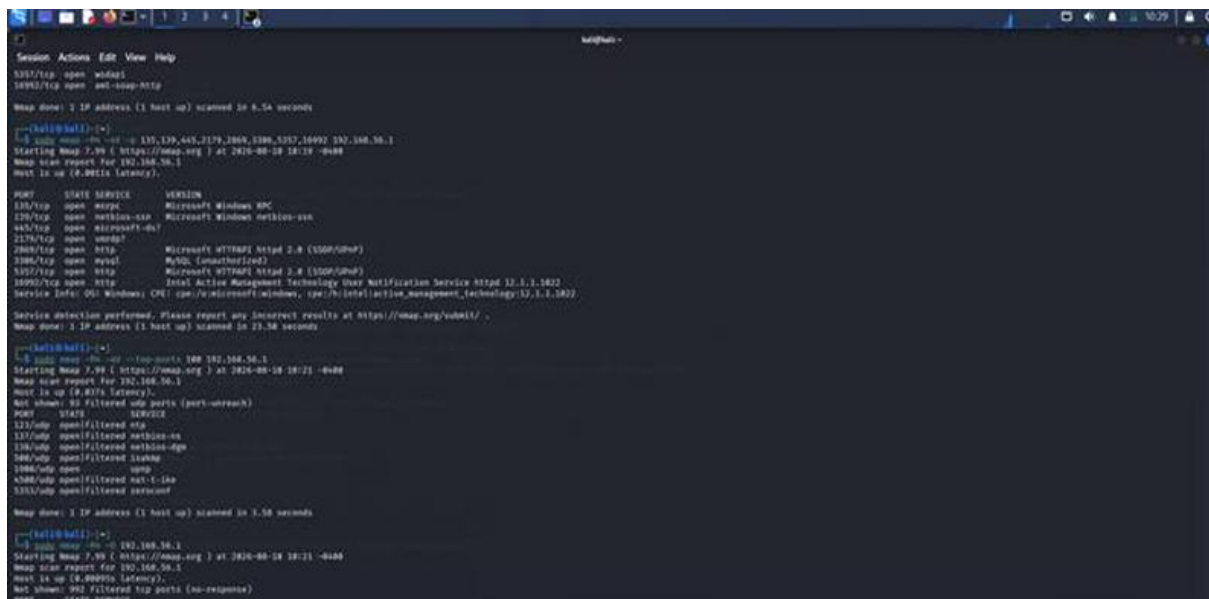
nmap -sV 192.168.56.1

Step 4: OS Fingerprinting

sudo nmap -O 192.168.56.

Step 5: Default Script Scan

nmap -sC -sV 192.168.56.1



```
Session Actions Edit View Help
5357/tcp open  wsdapi
10992/tcp open  anti-snap-http

Nmap done: 1 IP address (1 host up) scanned in 6.54 seconds

--(kali@kali)~
[~] sudo nmap -p- -sV -P- -- 135,139,445,2179,2869,3306,5357,10992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).
PORT      STATE SERVICE        VERSION
135/tcp   open  murpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?  Microsoft Windows netbios-ssn
2179/tcp  open  vncpp?        Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
2869/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp  open  mysql         MySQL (unauthorized)
5357/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
10992/tcp open  http          Intel Active Management Technology User Notification Service httpd 12.1.1.1822
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1822

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.34 seconds

--(kali@kali)~
[~] sudo nmap -p- -sV --top-ports 1000 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  murpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
2179/tcp  open  vncpp?
2869/tcp  open  http
3306/tcp  open  mysql
5357/tcp  open  http
10992/tcp open  http
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1822

Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds

--(kali@kali)~
[~] sudo nmap -p- -sV --top-ports 1000 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 10:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.000000s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
```

Step 6: Management-Service Scan

nmap -p 22,23,80,443,161,162 192.168.56.1

```
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds

--kali@kali:~$-
$ sudo nmap -p 22,23,80,443,161,162 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-20 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).
Not shown: 655 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
143/tcp   open  microsoft-sql
170/tcp   open  smtp
180/tcp   open  cifs
2380/tcp  open  mysql
3307/tcp  open  mysql
3309/tcp  open  mysql
3392/tcp  open  smtp-http

Nmap detection results may be unreliable because we could not find at least 1 open and 1 closed port.
Device type: VFP adapter [intel] (general purpose)
Running (Nmap) OS detection: OSes rejected: GNU, Oracle Solaris (OS), Suse (SUSE), IBM (OS)
OS CPE: cpe:/o:oracle:virtuallbox rpm:/s:many-generators:all:linux (cpe:/o:oracle:virtuallbox rpm:/s:many-generators:all:linux)
Aggression by guess: not enough valid gateway (N/A), Oracle Virtuallbox Ssls not bridge (N/A), GNU user mode network gateway (N/A)
No exact OS matches for host (fast scan) (see nmap-1080).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 14.18 seconds

--kali@kali:~$-
$ sudo nmap -p 22,23,80,443,161,162,3307,3309,3392 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-20 18:22 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
143/tcp   open  microsoft-sql
170/tcp   open  smtp
180/tcp   open  cifs
2380/tcp  open  mysql
3307/tcp  open  mysql
3309/tcp  open  mysql
3392/tcp  open  smtp-http
3392/tcp  open  smtp-http

Nmap detection results may be unreliable because we could not find at least 1 open and 1 closed port.
Device type: VFP adapter [intel] (general purpose)
Running (Nmap) OS detection: OSes rejected: GNU, Oracle Solaris (OS), Suse (SUSE), IBM (OS)
OS CPE: cpe:/o:oracle:virtuallbox rpm:/s:many-generators:all:linux (cpe:/o:oracle:virtuallbox rpm:/s:many-generators:all:linux)
Aggression by guess: not enough valid gateway (N/A), Oracle Virtuallbox Ssls not bridge (N/A), GNU user mode network gateway (N/A)
No exact OS matches for host (fast scan) (see nmap-1080).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 14.18 seconds
```

Management Interface Analysis

SSH — Port 22

SSH provides secure remote administration.

Security concerns

- Weak passwords
- Exposed authentication service
- Vulnerable SSH implementation
- Excessive administrative access

```
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 8.38 seconds

--kali@kali:~$-
$ sudo nmap -p 22,23,80,443,161,162,3307,3309,3392 --script http-title,http-headers,http-methods 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-20 18:26 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).
PORT      STATE SERVICE
22/tcp    open  ssh
3307/tcp  open  mysql
Nmap done: 1 IP address (1 host up) scanned in 8.62 seconds

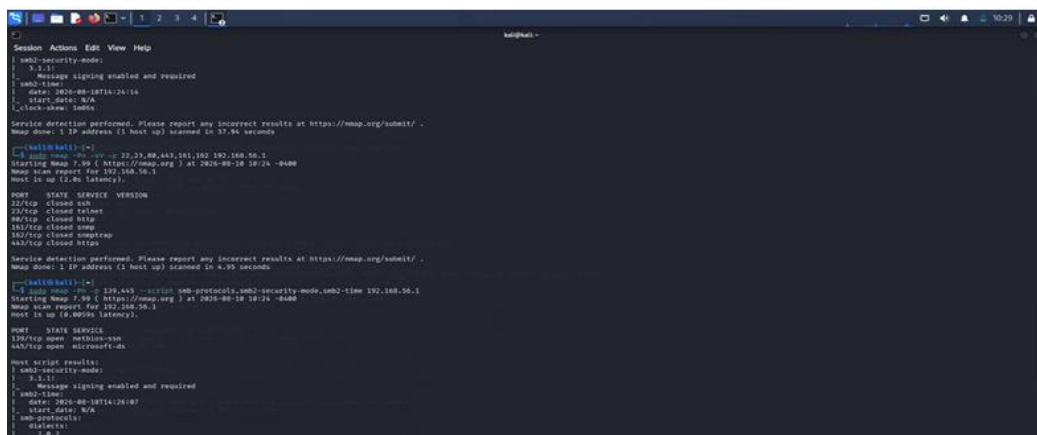
--kali@kali:~$-
$ sudo nmap -p 22,23,80,443,161,162,3307,3309,3392 --script upnp-info 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-20 18:26 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).
PORT      STATE SERVICE
2380/tcp  open  upnp
Nmap done: 1 IP address (1 host up) scanned in 3.84 seconds

--kali@kali:~$-
$ sudo nmap -p 22,23,80,443,161,162,3307,3309,3392 --script http-title,http-headers,http-methods 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-20 18:27 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).
PORT      STATE SERVICE
2380/tcp  open  upnp
Nmap done: 1 IP address (1 host up) scanned in 8.63 seconds
```

Recommendation

- Use key-based authentication.
- Disable password authentication where practical.
- Restrict SSH to management hosts.
- Use MFA where supported.

Risk: Medium if restricted; High if unnecessarily exposed.



```
Session Actions Edit View Help
nmap2-security-mode:
  3.1.1
  Message signing enabled and required
nmap2-time:
  date: 2020-08-18T14:26:16
  start-date: N/A
  clock-show: 1000s
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.96 seconds

[~]kali@kali:~$ nmap -sV -p- -iL ip.txt -oX nmap.xml -oR nmap.txt 192.168.56.1
Starting Nmap 7.90 ( https://nmap.org ) at 2020-08-18 10:26 -0400
Nmap scan report for 192.168.56.1
Host is up (2.40s latency).

PORT      STATE SERVICE VERSION
22/tcp    closed ssh
23/tcp    closed telnet
80/tcp    closed http
135/tcp   closed msrpc
136/tcp   closed msrpc
137/tcp   closed msrpc
138/tcp   closed msrpc
139/tcp   closed msrpc
443/tcp   closed https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.95 seconds

[~]kali@kali:~$ nmap -sV -p- -iL ip.txt -oX nmap.xml -oR nmap.txt 192.168.56.1
Starting Nmap 7.90 ( https://nmap.org ) at 2020-08-18 10:26 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).

PORT      STATE SERVICE
138/tcp   open  netbios-ssn
443/tcp   open  microsoft-ds

Nmap script results:
nmap2-security-mode:
  3.1.1
  Message signing enabled and required
nmap2-time:
  date: 2020-08-18T14:26:07
  start-date: N/A
  clock-show: 1000s
nmap2-protocols:
  2.0.2
```

Telnet — Port 23

Telnet is an insecure remote-management protocol.

Authentication information can be transmitted without modern encryption.

Recommendation

Disable Telnet.

Use:

SSH instead of Telnet

Risk: High.

HTTP — Port 80

HTTP may provide a router/gateway management interface.

The major concern is that management credentials and administrative information may be exposed if authentication is not protected properly.

Recommendation

- Disable HTTP management if unnecessary.
- Redirect or replace with HTTPS where supported.
- Restrict access to the management network.

Risk: High if it exposes administration without adequate protection.

HTTPS — Port 443

HTTPS is preferable to HTTP because it provides encrypted communication.

However, security still depends on:

- Strong authentication
- Valid TLS configuration
- Current software
- Access restrictions
- No default credentials

Risk: Medium when properly configured; High if vulnerable or broadly exposed.

SNMP — Port 161

SNMP is used for network monitoring and management.

Older SNMP configurations may use weak community strings.

Potential information exposure includes:

- Device information
- Interface information
- Network configuration
- System information

Recommendation

- Prefer SNMPv3.
- Disable SNMP if not required.
- Restrict SNMP access to monitoring servers.
- Use strong authentication/privacy settings.

Risk: Medium/High depending on configuration.

If unnecessary management services are exposed, attackers gain additional opportunities for reconnaissance and unauthorized access.

How Attackers Could Abuse Management Services

2. Credential Attacks

Exposed SSH or Telnet may attract authentication attacks.

3. Information Gathering

SNMP may reveal useful network information if improperly configured.

4. Management Interface Attacks

A vulnerable HTTP/HTTPS management interface could potentially expose administrative functionality.

5. Network Compromise

Compromising a gateway is particularly serious because it can provide a privileged position within the network

Security Controls

1. Disable Unnecessary Services

If a service has no business requirement:

Disable it.

2. Restrict Management Access

Management interfaces should be accessible only from trusted administrator systems.

3. Disable Telnet

Replace:

Telnet → SSH

4. Secure SSH

Use:

- Strong credentials
- SSH keys
- MFA
- IP restrictions
- Account monitoring

5. Secure SNMP

Prefer:

SNMPv3

over older versions when supported.

6. Secure Web Management

Use:

HTTPS

instead of HTTP.

Change default credentials immediately

7. Network Segmentation

Separate:

- User network
- Server network
- Management network
- Guest network

8. Logging and Monitoring

Monitor:

- SSH login attempts
- Administrative logins
- Configuration changes
- SNMP access
- Web management access

Overall Conclusion

Nmap enumeration is an important first phase of an authorized penetration test because it establishes the target's externally visible attack surface.